aapptec

Technical Support
Information Bulletin 1085

Removing Trifluoroacetic Acid (TFA) From Peptides

TFA/HCl Exchange in Water

1. Dissolve the peptide in 100 mM HCl.
2. Allow the solution to stand at room temperature for 1 minute.
3. Freeze the solution in liquid nitrogen.
4. Lyophylize the frozen solution to obtain the peptide hydrochloride salt.

   This process may need to be repeated several times to reduce TFA content to acceptable levels.

TFA/HCl Exchange in Acetonitrile[1]

1. Dry acetonitrile over molecular sieves.
2. Saturate the acetonitrile with gaseous HCl
3. Dissolve the peptide in HCl-saturated acetonitrile (1 mg peptide/ml).
4. Stir at room temperature for 5 minutes.
5. Evaporate at 40 ˚C under vacuum on a rotary evaporator.
6. Dissolve the residue in water and lyophilize.

   The peptide will be approximately 98 mol% Cl^-

TFA/Acetate Exchange

1. Prepare a small column (10-fold to 50-fold excess of anion sites in the column relative to anion sites in the peptide) of strong anion exchange resin.
2. Elute the column with a 1M solution of sodium acetate.
3. Wash the column with distilled water to remove the excess sodium acetate.
4. Dissolve the peptide in distilled water and apply it to the column.

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[1] K. Sikora, D. Neubauer, M. Jaśkiewicz, W. Kamysz, *Int. J. Pept. Res. Ther.,* **2018,** *24*, 265-279. DOI: 10.1007/s10989-017-9611-7

5. Elute the column with distilled water and collect the fractions containing the peptide.
6. Lyophylize the combined peptide containing fractions to obtain the peptide acetate salt.